10.2.4 Cost of a simple QR algorithm ¶

The QR algorithms that we have discussed incur the following approximate costs per iteration for an $m \times m$ Hermitian matrix.

- $A \rightarrow QR$ (QR factorization): $\frac{4}{3}m^3$ flops.
- $A := RQ$. A naive implementation would take advantage of R being upper triangular, but not of the fact that A will again be Hermitian, at a cost of m^3 flops. If one also takes advantage of the fact that A is Hermitian, that cost is reduced to $\frac{1}{2}m^3$ flops.
- If the eigenvectors are desired (which is usually the case), the update $V := VQ$ requires an addition $2m^3$ flops.

Any other costs, like shifting the matrix, are inconsequential.

Thus, the cost, per iteration, equals approximately $(\frac{4}{3} + \frac{1}{2})m^3 = \frac{11}{6}m^3$ flops if only the eigenvalues are to be computed. If the eigenvectors are also required, then the cost increases by $2m^3$ flops to become $\frac{23}{6}m^3$ flops.

Let us now consider adding deflation. The rule of thumb is that it takes a few iterations per eigenvalue that is found. Let's say K iterations are needed. Every time an eigenvalue is found, the problem deflates, decreasing in size by one. The cost then becomes

$$\sum_{i=m}^{1} K\frac{23}{6}i^3 \approx K\frac{23}{6}\int_0^m x^3 dx = K\frac{23}{6}\frac{1}{4}x^4|_0^m = K\frac{23}{24}m^4.$$

The bottom line is that the computation requires $O(m^4)$ flops. All other factorizations we have encountered so far require at most $O(m^3)$ flops. Generally $O(m^4)$ is considered prohibitively expensive. We need to do better!